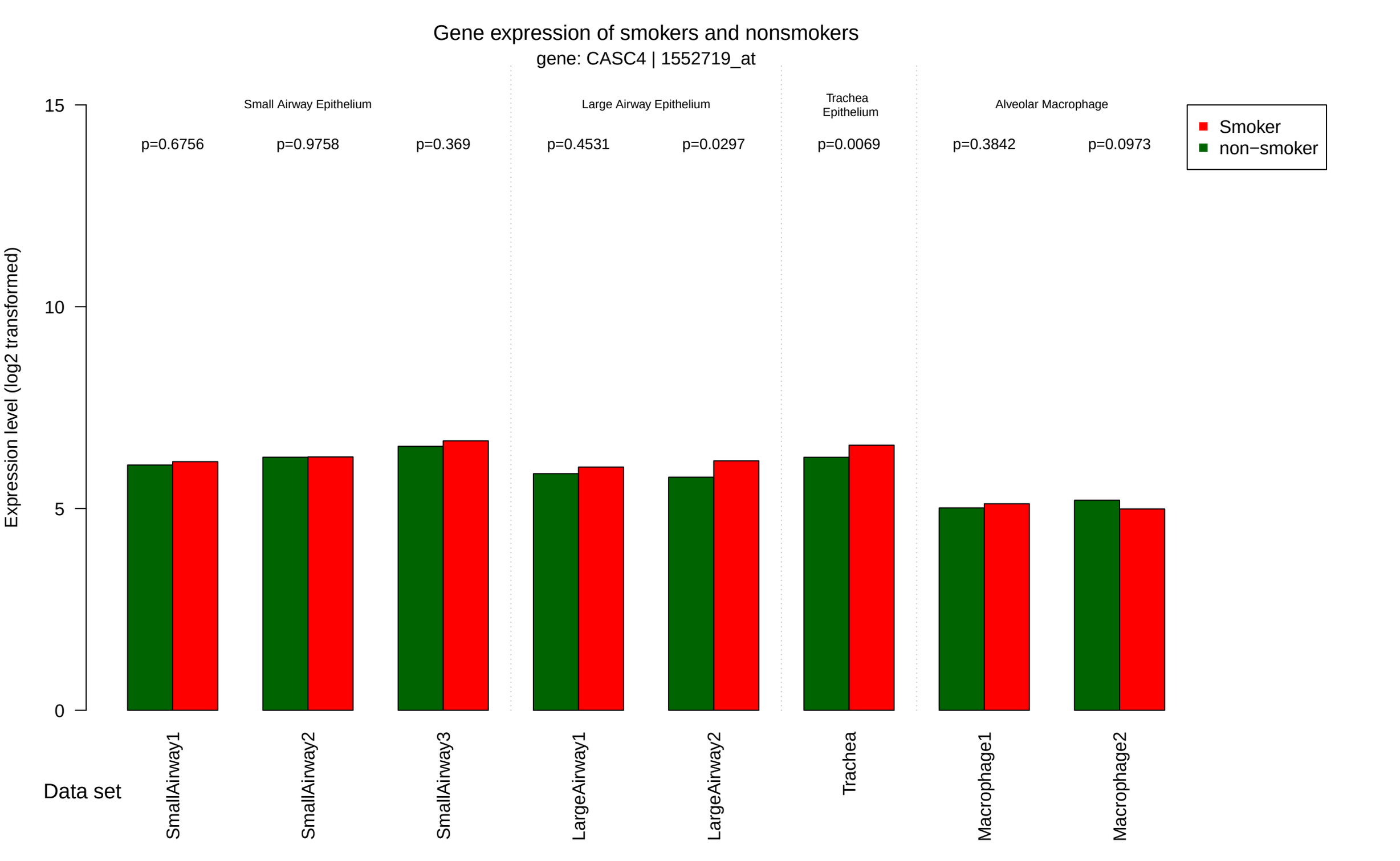
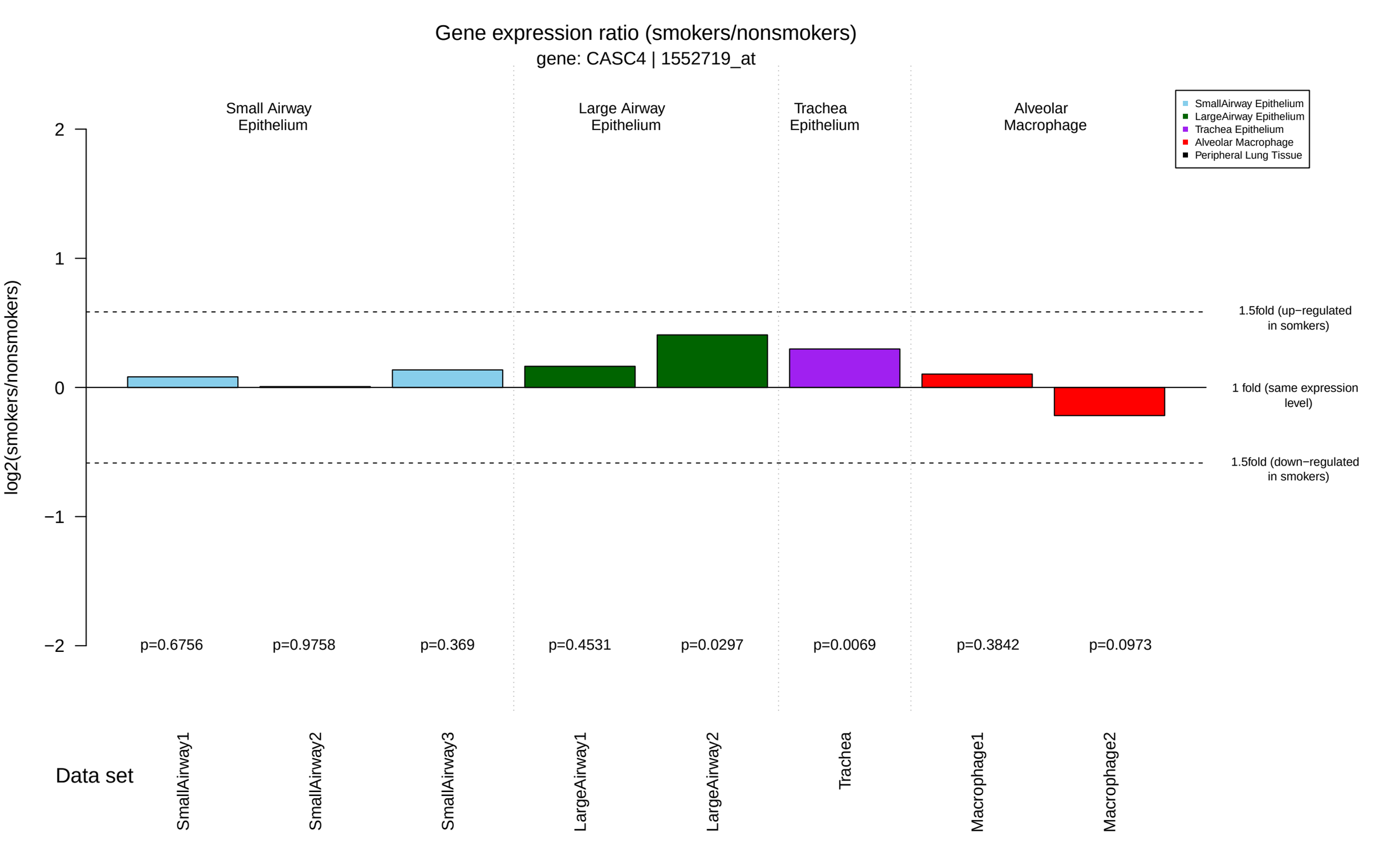
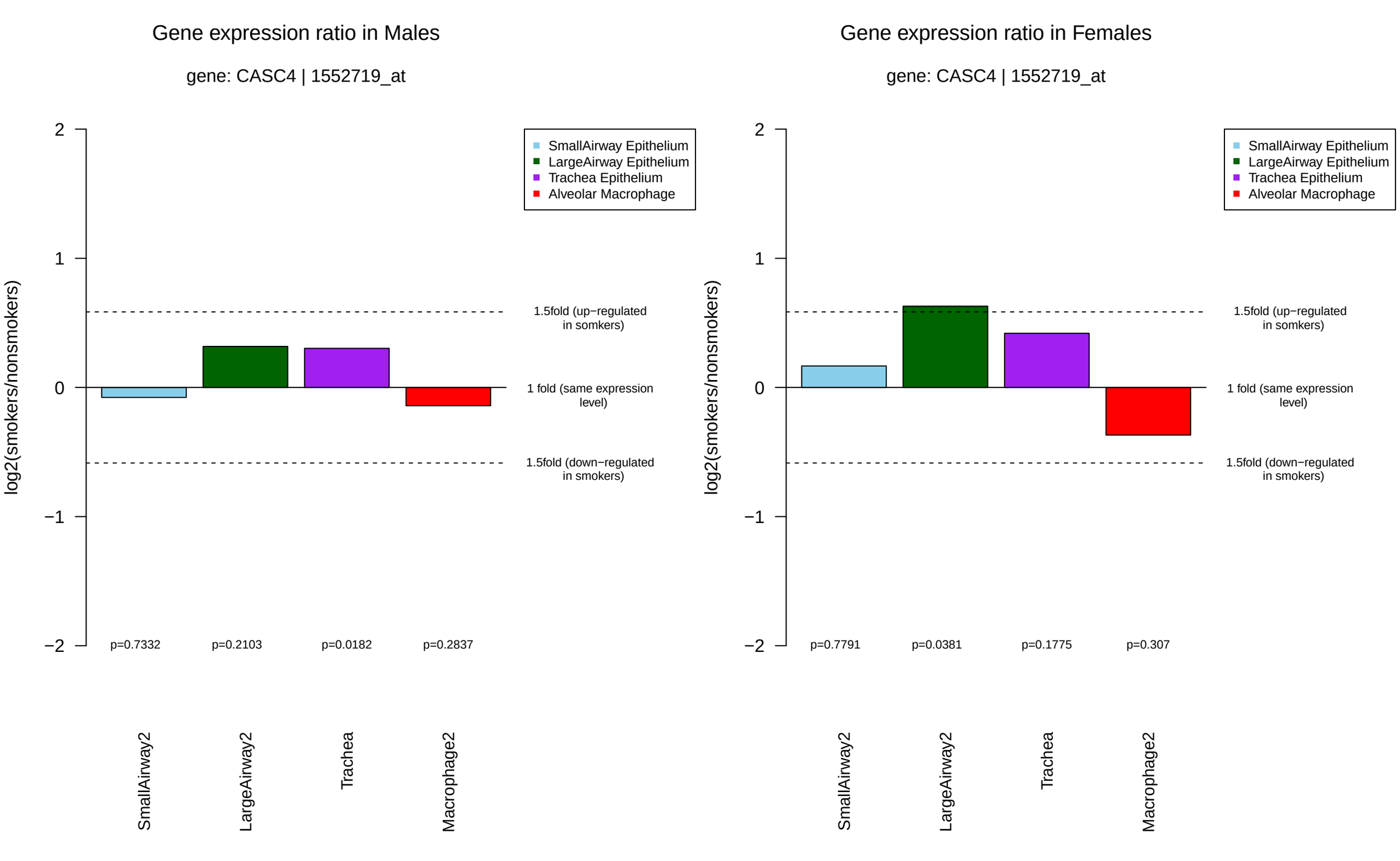
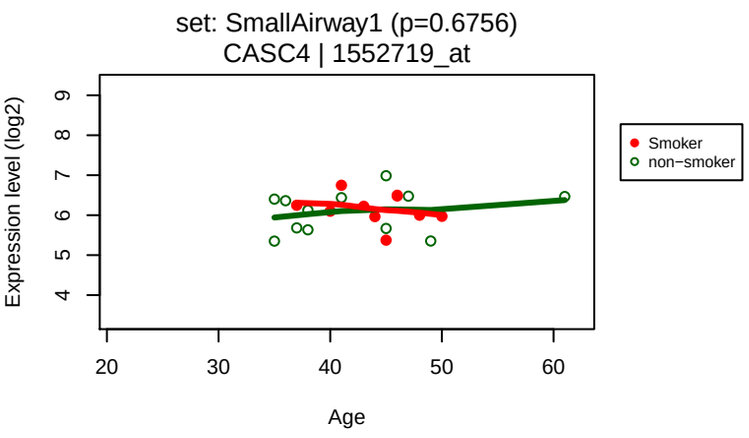
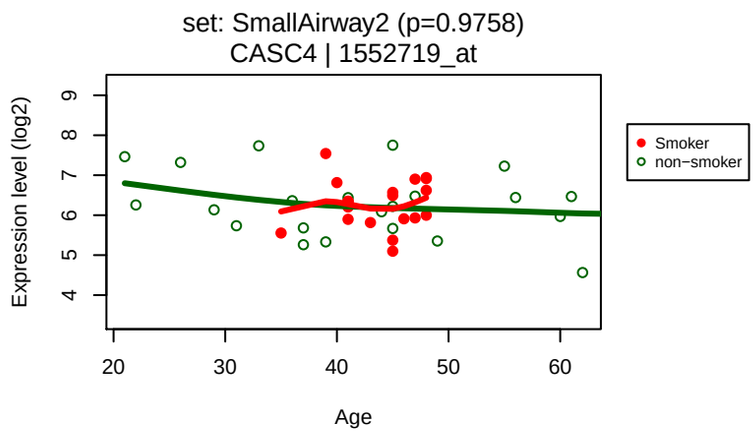
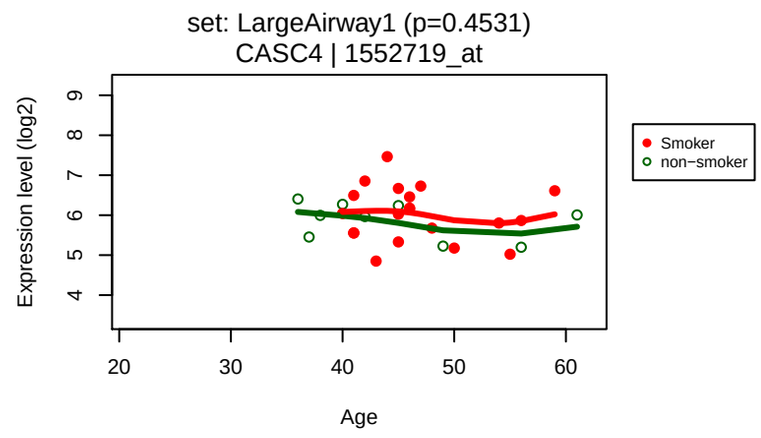

Gene expression of smokers and nonsmokers
gene: CASC4 | 1552719_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6756
p=0.9758
p=0.369
p=0.4531
p=0.0297
p=0.0069
p=0.3842
p=0.0973
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

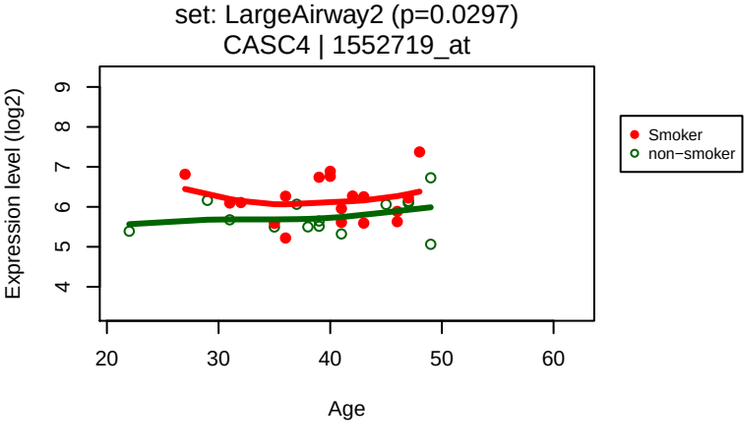

Gene expression ratio (smokers/nonsmokers)
gene: CASC4 | 1552719_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6756
p=0.9758
p=0.369
p=0.4531
p=0.0297
p=0.0069
p=0.3842
p=0.0973
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

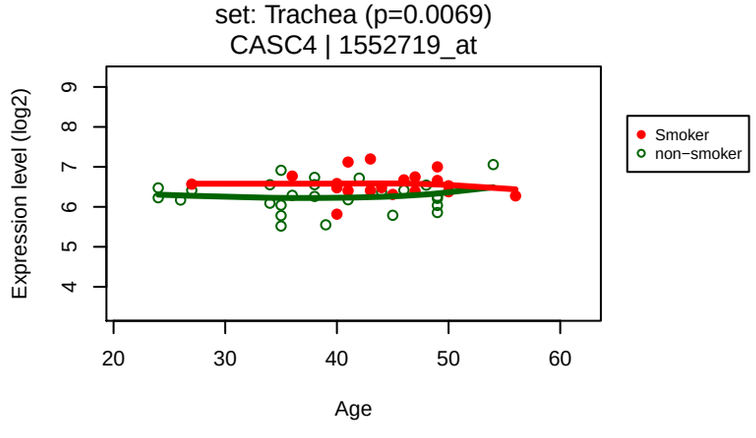

Gene expression ratio in Males
gene: CASC4 | 1552719_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7332
p=0.2103
p=0.0182
p=0.2837
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CASC4 | 1552719_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7791
p=0.0381
p=0.1775
p=0.307
SmallAirway2
LargeAirway2
Trachea
Macrophage2

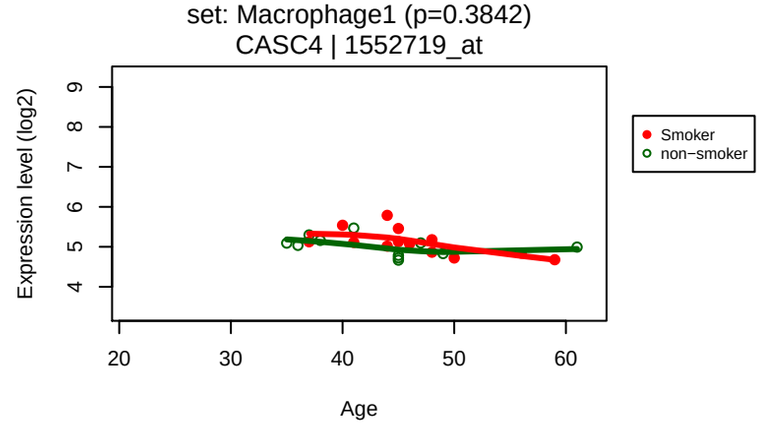

set: SmallAirway1 (p=0.6756)
CASC4 | 1552719_at
Expression level (log2)
Age
Smoker
non−smoker

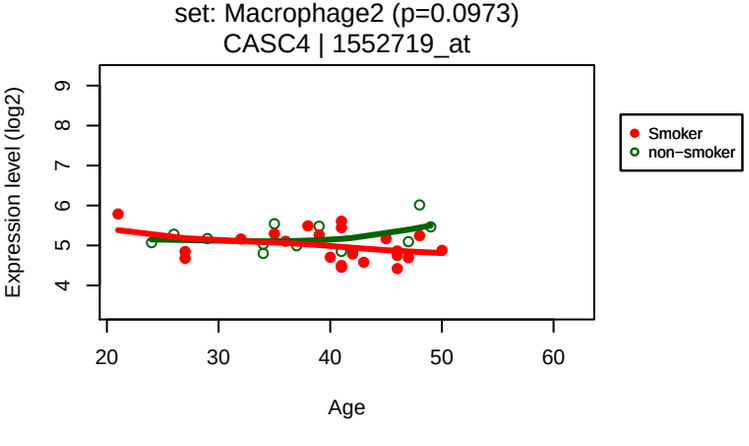

set: SmallAirway2 (p=0.9758)
CASC4 | 1552719_at
Expression level (log2)
Age
Smoker
non−smoker

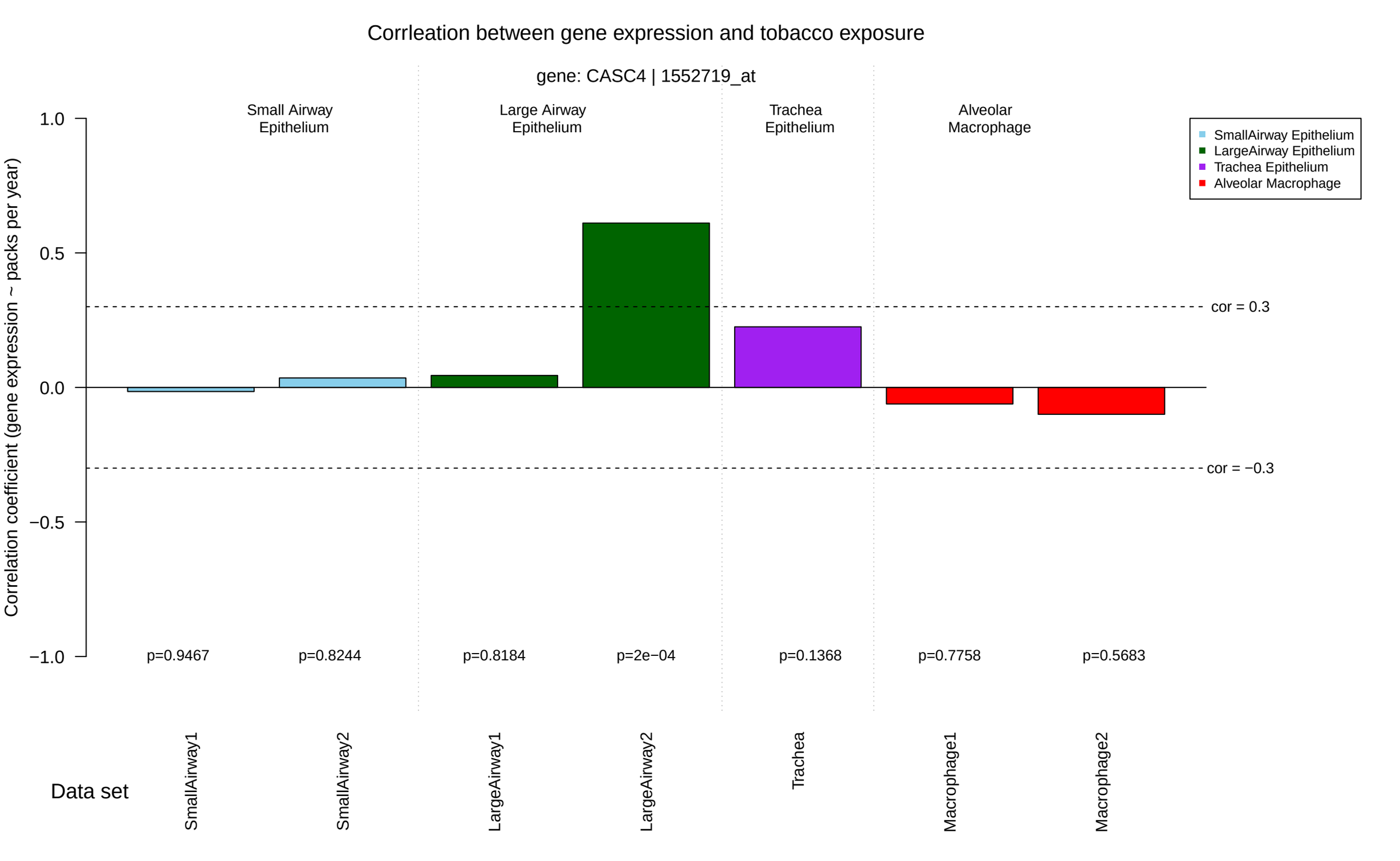

set: LargeAirway1 (p=0.4531)
CASC4 | 1552719_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0297)
CASC4 | 1552719_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0069)
CASC4 | 1552719_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3842)
CASC4 | 1552719_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0973)
CASC4 | 1552719_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CASC4 | 1552719_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9467
p=0.8244
p=0.8184
p=2e−04
p=0.1368
p=0.7758
p=0.5683
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2